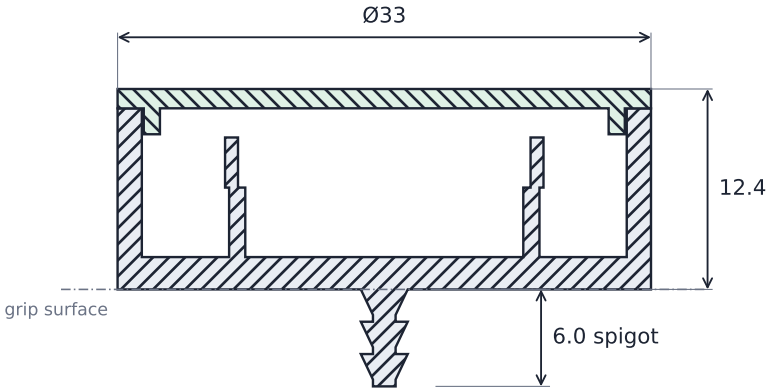
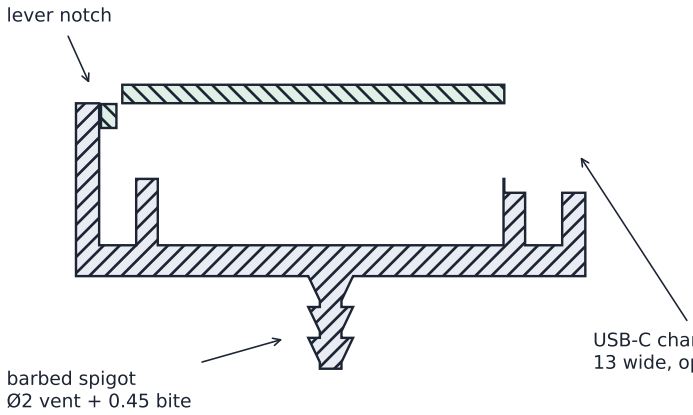
Print drawing: putter pod cup and lid

For the print shop. Print from the STL files; this sheet is the reference for checking them.

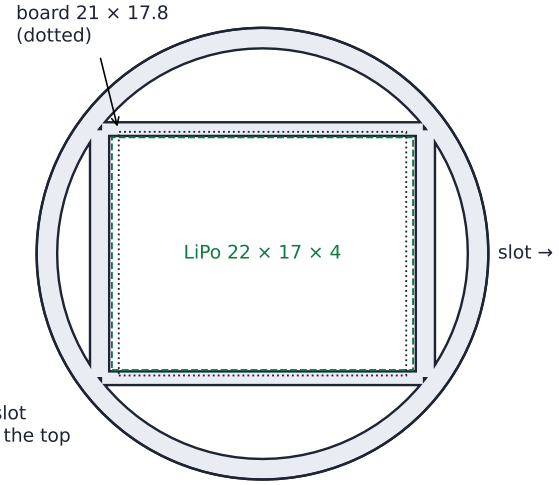
Section A–A (across the rails), cup B with lid



Section B–B (along the board)



Plan, cut at battery height

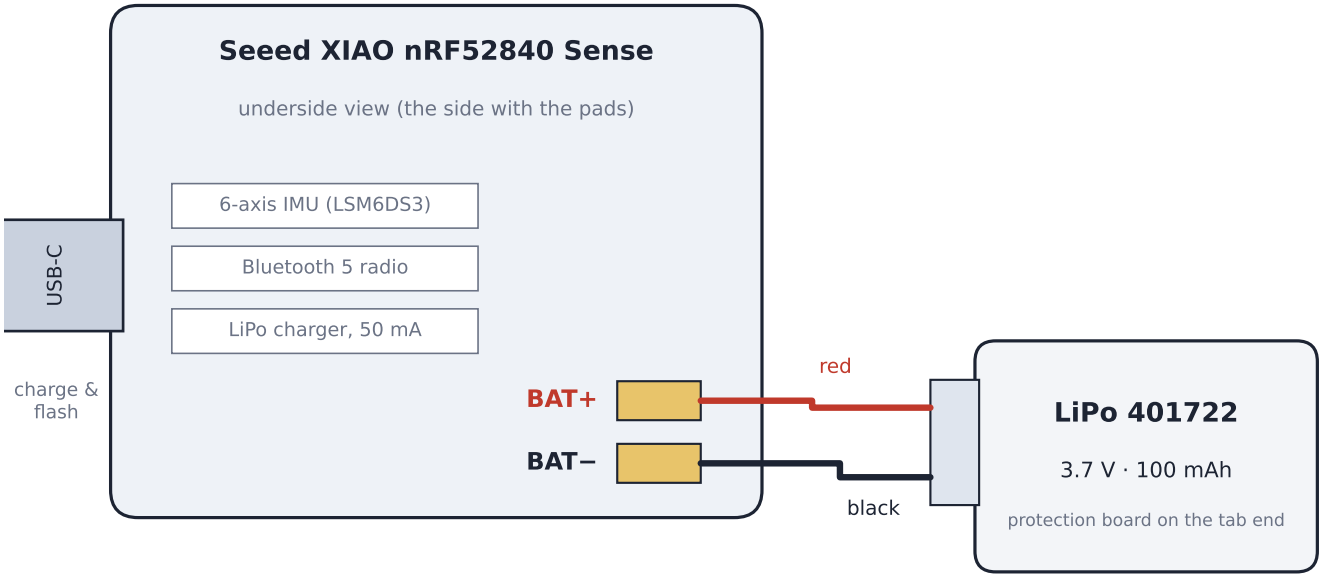


Material	PA12 nylon, SLS or MJF (e.g. JLC3DP, Shapeways, a local SLS bureau). Not resin: the barbs and lid lip crack.
Finish	As printed (grey or dyed black). No support marks on the lid lip or the spigot barbs.
Quantity	3 × cup B, 1 × cup A, 1 × cup D, 5 × lid (one spare). Cups A and D are for friends' putters.
Files	pod-cup-B-g33-v2.stl · pod-cup-A-g33-v2.stl · pod-cup-D-g33-v2.stl · pod-lid-g33.stl
Tolerances	±0.15 on the lid lip (it press-fits into the cup with 0.12 clearance). Everything else ±0.2.
Orientation	Printer's choice for SLS/MJF. For a test print on an FDM machine: cup flange-down with supports under the spigot, lid top-down, 0.2 layers, 100% infill.
Before ordering	Measure the putter (sheet 3, step 1) and regenerate the files if any number differs from the title block.

osolo putter pod, v1	Sheet 1 of 3
Ø33 / vent 2 / stop 8: to confirm	mm, 13 Nov 2026

Wiring: one battery, two wires

The motion sensor, Bluetooth and charger are all on the XIAO board. The only soldering is the battery.



about 25 mm each, keep the original battery leads if they reach

Polarity

Red to BAT+, black to BAT−. Reversed, it can kill the charger chip. Check the battery label before cutting anything: some cells use red and blue, or red and black on a connector.

No switch

The firmware sleeps when the putter is still and wakes on movement. Nothing to wire.

Charging

USB-C at 50 mA, 2 to 3 hours from flat. The small LED by the USB-C lights while it charges.

Kapton tape

Over the battery before the board goes on top, so the board's underside can't touch the cell.

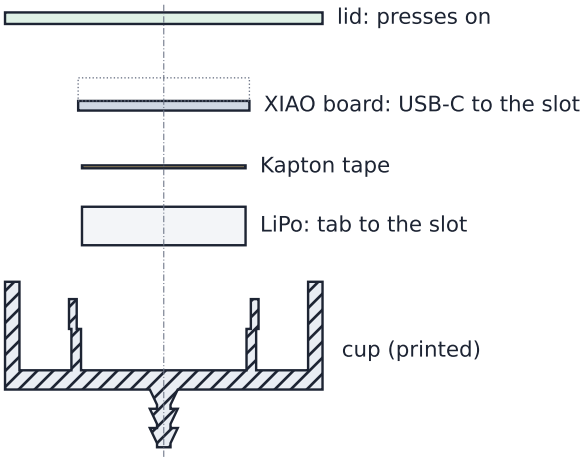
Never

Let the two battery leads touch, cut both at once, or solder with the iron above 350 °C on the cell side.

osolo putter pod, v1	Sheet 2 of 3
Ø33 / vent 2 / stop 8: to confirm	mm, 13 Nov 2026

Assembly, in order

About 30 minutes a pod once you have done one. Flash and test each board before it goes in.



1 Measure the putter

Calipers across the grip end (the butt, about 33), a drill-bit shank in the vent hole (the largest that slides in), a cocktail stick down the hole to the hard stop and mark the depth. Send the three numbers; the files are regenerated before printing.

2 Flash each board

Plug a board into the laptop, double-tap the tiny reset button by the USB-C; a drive called XIAO-SENSE appears. Drag osolo-putt-v1.uf2 onto it. It reboots. Tap the board on the desk: the green LED blinks.

3 Tin the pads

Board upside down in a helping hand. A dab of flux, then a little solder on BAT+ and BAT−. Fine tip, 320–340 °C, 2 seconds a pad.

4 Solder the battery

Trim and tin one battery lead at a time, never both cut together. Red to BAT+, black to BAT−. Unplug the USB first. Check with a multimeter: 3.7–4.2 V across the pads.

5 Test it alive

Plug USB back in: the charge LED lights. Unplug: tap the board, the green LED still blinks, so it runs on the battery. On the phone, the free nRF Connect app lists a device called osolo.

6 Fit it

Battery flat on the cup floor between the low rails, leads towards the slot. Kapton over it. Fold the wires into the gap at the slot end, board on the rail step, USB-C facing the slot.

7 Close and fit to the putter

Press the lid on (notch opposite the slot). Push the spigot into the grip's vent hole with a twist until the flange sits flat. If it's loose, a 33 mm ring of 1 mm VHB tape under the flange.

Tools Fine-tip soldering iron (temperature controlled), thin solder (0.5–0.8 mm), flux pen, helping hand, flush cutters, multimeter, Kapton tape, digital calipers, 1 mm VHB tape (optional), USB-C data cable.

osolo putter pod, v1	Sheet 3 of 3
Ø33 / vent 2 / stop 8: to confirm	mm, 13 Nov 2026